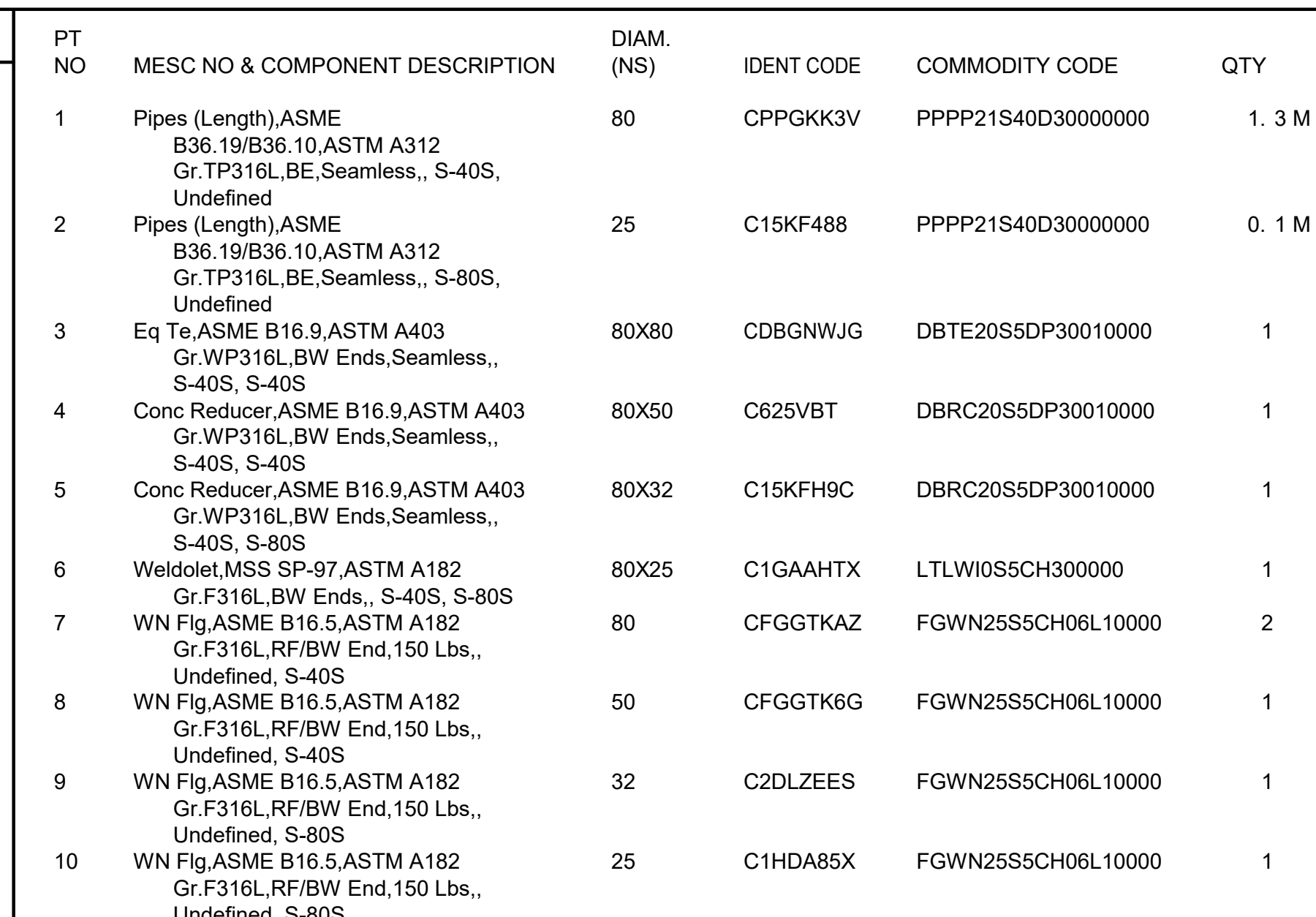




PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
11	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	80	C28Z7LJH	GKSW21100G0OL100085400	2
12	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	50	C28Z7LJC	GKSW21100G0OL100085400	2
13	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	32	C2DLZEJA	GKSW21100G0OL100085400	1
14	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	25	C28Z7LJ4	GKSW21100G0OL100085400	2
15	SBit 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 180 mm Length	5/8	C10SN9DS	BTS222S00B010000	4
16	SBit 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 90 mm Length	5/8	C10SN9ES	BTS222S00B010000	8
17	SBit 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 80 mm Length	1/2	C10SN9DF	BTS222S00B010000	4
18	SBit 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 70 mm Length	1/2	C10SN9DE	BTS222S00B010000	8

VALVES / IN-LINE ITEMS						
19	DuPiChk Waf,API 594,ASTM A351 Gr.CF3M,RF or FF,150 Lbs., Undefined, Undefined	80	C2ETE54G	VCDD20S40301L10F02DR	1	
20	Bal Flg,FB,LPI,ISO 17292,ASTM A351 Gr.CF3M,RF,150 Lbs., Undefined, Undefined	50	C26WDTVJ	VBAH80S4030AL1UY077MLE	1	
21	Bal Flg,FB,LPI,ISO 17292,ASTM A182 Gr.F316L,RF,150 Lbs., Undefined, Undefined	25	C26WDTVf	VBAH80S5CH0AL1UY077MLE	1	

	<u>INSTRUMENTS</u>			
22	PIN444205	50		1
	<u>PIPE SUPPORTS</u>			
23	CR01 3" 100 300 SS	80	..	1
24	SP02 200	80	..	1
25	SB03 A H10 100 500 987	80	..	1
26	WG02 3" C	80	..	1

0	2026.06.22	ISSUE FOR CONSTRUCTION	MLZ	ZLF	
REV	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED

BANC PANDA3 PROJECT
蓝星安迪苏南京液体蛋氨酸装置技术改造项目

 TEN TECHNIP ENERGIES	T.EN Chemical Engineering (Tianjin) Co.,LTD 德希尼布化学工程（天津）有限公司	 Adisseo A Bluestar Company	Bluestar Adisseo Nanjing Co.,LTD 蓝星安迪苏南京有限公司
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	PIPING ISOMETRIC				
UNIT	AREA		INSU. THK	STRESS CAT	DOCUMENT CLASS
N400	N400P3A		30 mm		B

LINE NUMBER		MARIAN SHEET
80 AMSS2 N444201 A2DA10 PP		01
PID NO.	SHEET	REV
2069N-N400-PID-0021-N444	1 OF 5	0

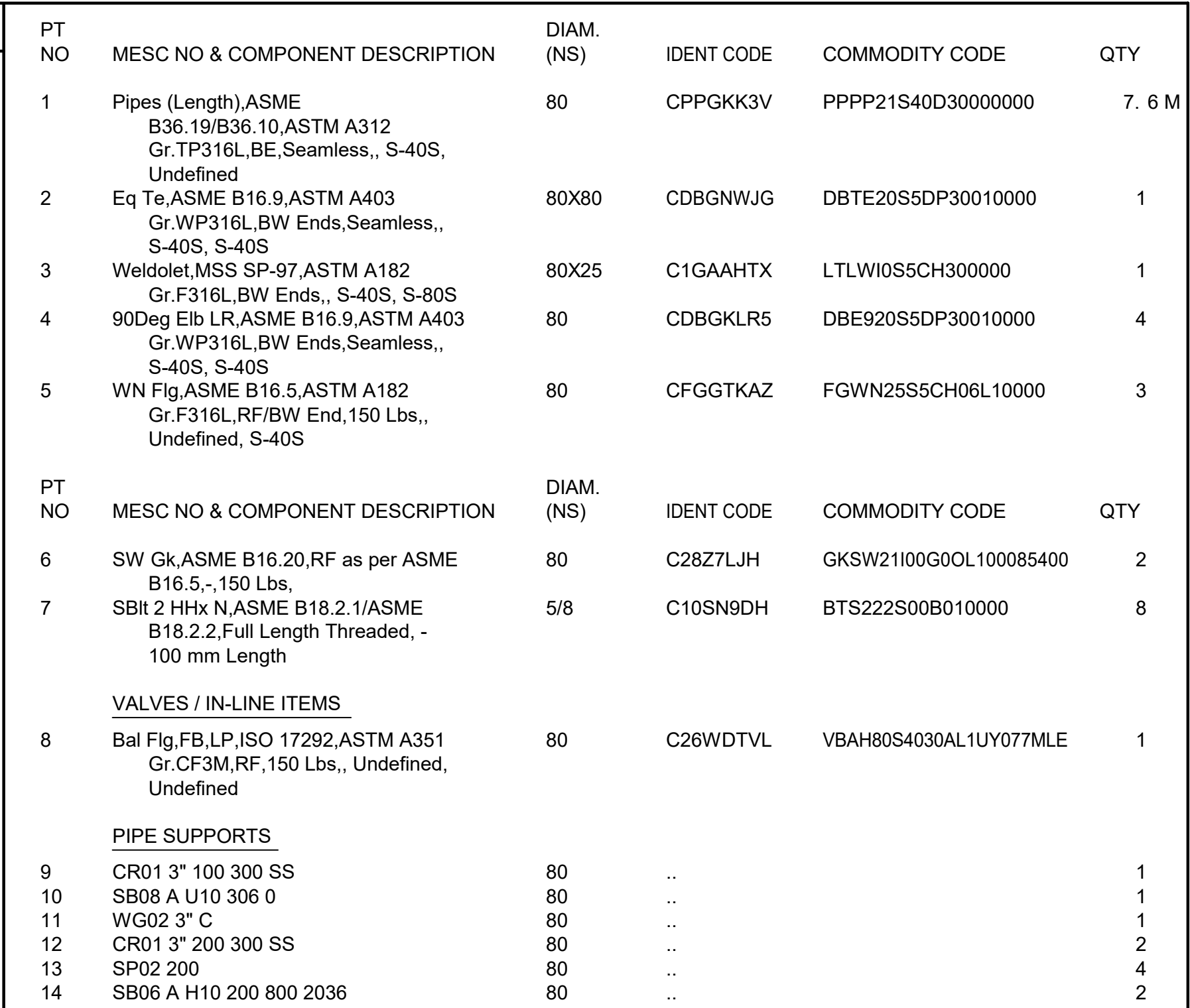
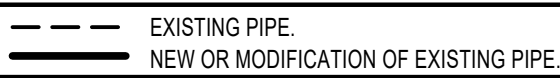
N400P3A-AMSS2-N444201-01

N400P3A-AMSS2-N444201-01

NOTES: 1. FOR HOLD ITEMS, TPT PIPING FIELD ENGINEER TO CONFIRM FINAL DIMENSION AT SITE.
2. PIPE ROUTING AND SUPPORTS OF LINES DN40 AND SMALLER (NO STRESS ANALYSED) ARE TO BE CONFIRMED AT SITE BY PIPING SUB-CONTRACTOR.
3. FOR STRESS ANALYSED LINES, NO CHANGES ALLOWED WITHOUT WRITTEN APPROVAL FROM PIPE STRESS ENGINEER.

EXTRACTION DATE: Today's Date (Locale Sensitive format) : 6/17/2026 9:53:27 AMToday's Date (Your format) : 17/06/26Today's Date (ISO8601 Small) : 2026-06-17 09:53:27

TPC S3D BANC2\BANC2\N400\2069N\Piping\Pipelines\2069N-N400-PID-0021-N444

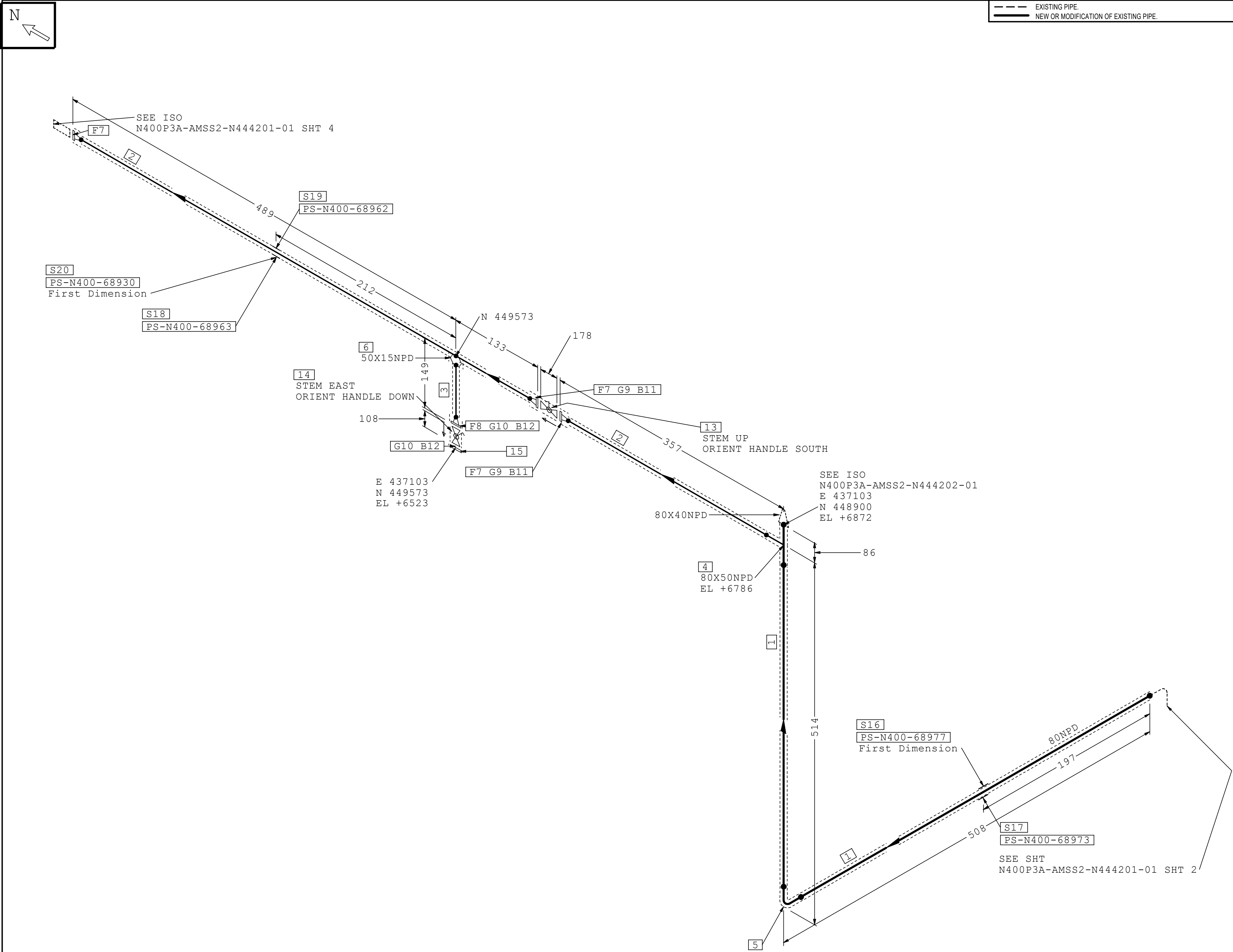


0	2026.06.22	ISSUE FOR CONSTRUCTION		MLZ	ZLF
REV	DATE	DESCRIPTION		DRAWN	CHECKED APPROVED
BANC PANDA3 PROJECT 蓝星安迪苏南京液体蛋氨酸装置技术改造项目					
 TECHNIP ENERGIES		T.EN Chemical Engineering (Tianjin) Co.,LTD 德希尼布化学工程（天津）有限公司		 Bluestar Adisseo Nanjing Co.,LTD 蓝星安迪苏南京有限公司	
	PIPING ISOMETRIC				
UNIT	AREA		INSU. THK	STRESS CAT	DOCUMENT CLASS
N400	N400P3A		30 mm		B
LINE NUMBER					MARIAN SHEET
80 AMSS2 N444201 A2DA10 PP					01
PID NO.				SHEET	REV
2069N-N400-PID-0021-N444				2 OF 5	0

TPC S3D BANC2\BANC2\N400\2069N\Piping\Pipelines\2069N-N400-PID-0021-N444

N400P3A-AMSS2-N444201-01

N400P3A-AMSS2-N444201-01



PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
1	Pipes (Length),ASME B36.19/B36.10,ASTM A312 Gr.TP316L,BE,Seamless,, S-40S, Undefined	80	CPPGKK3V	PPPP21S40D30000000	0. 8 M
2	Pipes (Length),ASME B36.19/B36.10,ASTM A312 Gr.TP316L,BE,Seamless,, S-40S, Undefined	50	CPPGKK29	PPPP21S40D30000000	0. 8 M
3	Pipes (Length),ASME B36.19/B36.10,ASTM A312 Gr.TP316L,BE,Seamless,, S-80S, Undefined	15	C1GA7AUA	PPPP21S40D30000000	0. 1 M
4	Red Te,ASME B16.9,ASTM A403 Gr.WP316L,BW Ends,Seamless,, S-40S, S-40S	80X50	CDBGS59M	DBTR20S5DP30010000	1
5	90Deg Elb LR,ASME B16.9,ASTM A403 Gr.WP316L,BW Ends,Seamless,, S-40S, S-40S	80	CDBGKLR5	DBE920S5DP30010000	1
6	Weldolet,MSS SP-97,ASTM A182 Gr.F316L,BW Ends,, S-40S, S-80S	50X15	C1GAABZ2	LTLW10S5CH300000	1
7	WN Fig,ASME B16.5,ASTM A182 Gr.F316L,RF/BW End,150 Lbs., Undefined, S-40S	50	CFGGTK6G	FGWN25S5CH06L10000	3
8	WN Fig,ASME B16.5,ASTM A182 Gr.F316L,RF/BW End,150 Lbs., Undefined, S-80S	15	CXSMBEX	FGWN25S5CH06L10000	1

PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
9	SW Gk,ASME B16.20,RF as per ASME B16.5-,150 Lbs,	50	C28Z7LJC	GKSW21100G0L100085400	2
10	SW Gk,ASME B16.20,RF as per ASME B16.5-,150 Lbs,	15	C28Z7LJ2	GKSW21100G0L100085400	2
11	SBIt 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 90 mm Length	5/8	C10SN9ES	BTS222S00B010000	8
12	SBIt 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 70 mm Length	1/2	C10SN9DE	BTS222S00B010000	8

VALVES / IN-LINE ITEMS					
13	Bal Fig,FB,LP,ISO 17292,ASTM A351 Gr.CF3M,RF,150 Lbs., Undefined, Undefined	50	C26WDTVJ	VBAH80S4030AL1UY077MLE	1
14	Bal Fig,FB,LP,ISO 17292,ASTM A182 Gr.F316L,RF,150 Lbs., Undefined, Undefined	15	C26WDTV D	VBAH80S5CH0AL1UY077MLE	1
15	Bld Fig,ASME B16.5,ASTM A182 Gr.F316L,RF,150 Lbs., Undefined, Undefined	15	CFG66DP6	FGBL25S5CH02L10000	1

PIPE SUPPORTS					
16	SB08 A H12 341 0	80	..		1
17	CR01 3" 100 300 SS	80	..		1
18	CR01 2" 100 300 SS	50	..		1
19	WG02 2" C	50	..		1
20	SB04 A H10 150 400 634	50	..		1

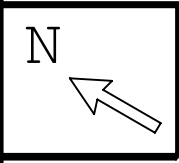
0	2026.06.22	ISSUE FOR CONSTRUCTION	MLZ	ZLF	
REV	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED

BANC PANDA3 PROJECT
蓝星安迪苏南京液体蛋氨酸装置技术改造项目

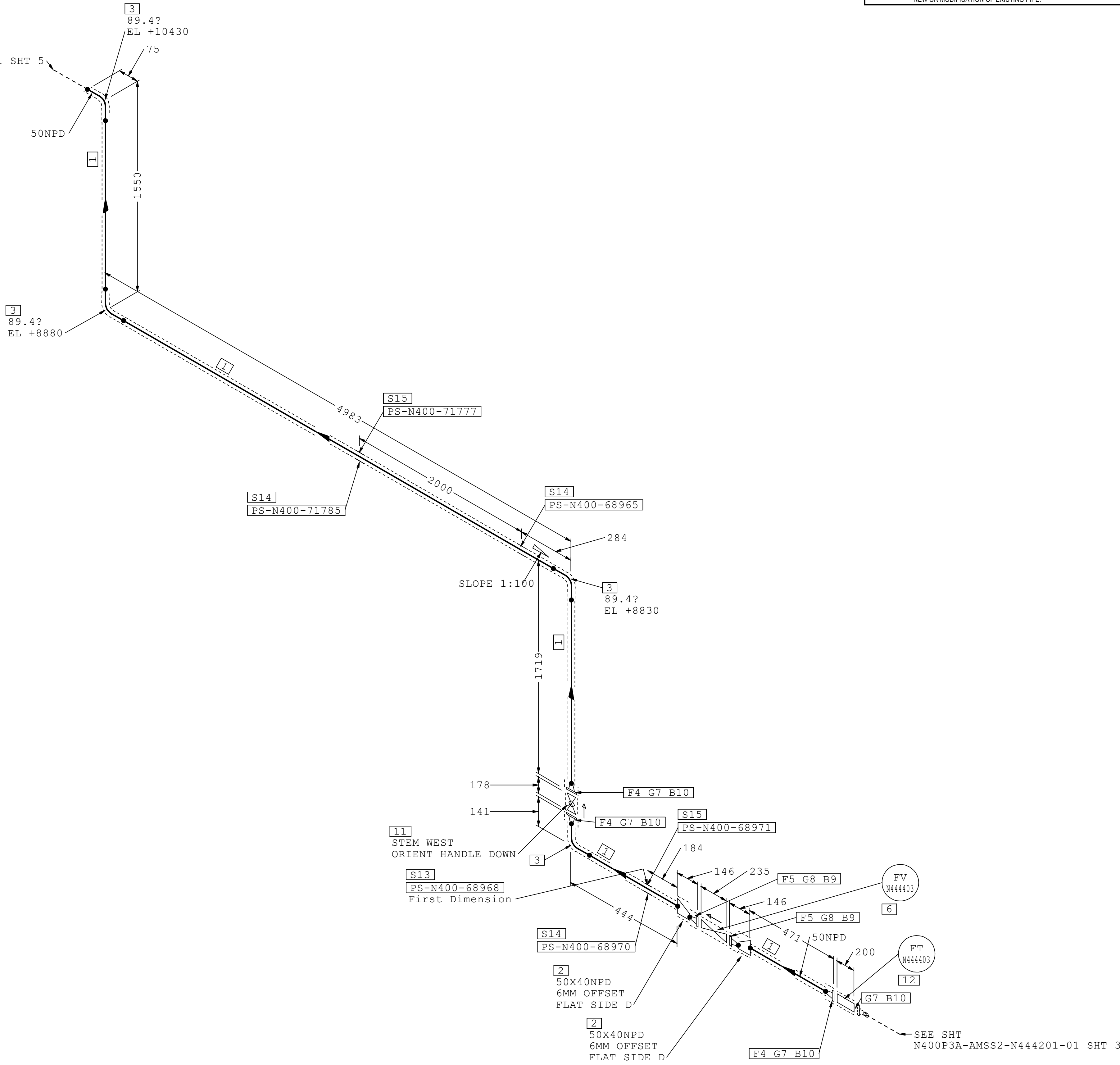
T.EN Chemical Engineering (Tianjin) Co.,LTD
德希尼布化学工程（天津）有限公司

Bluestar Adisseo Nanjing Co.,LTD
蓝星安迪苏南京有限公司

PIPING ISOMETRIC					
UNIT	AREA		INSU. THK	STRESS CAT	DOCUMENT CLASS
N400	N400P3A		30 mm		B
LINE NUMBER					MARIAN SHEET
80 AMSS2 N444201 A2DA10 PP					01
PID NO.				SHEET	REV
2069N-N400-PID-0021-N444				3 OF 5	0



SEE ISO
N400P3A-AMSS2-N444201-01 SHT 5



NOTES: 1. FOR HOLD ITEMS, TPT PIPING FIELD ENGINEER TO CONFIRM FINAL DIMENSION AT SITE.
2. PIPE ROUTING AND SUPPORTS OF LINES DN40 AND SMALLER (NO STRESS ANALYSED) ARE TO BE CONFIRMED AT SITE BY PIPING SUB-CONTRACTOR.
3. FOR STRESS ANALYSED LINES, NO CHANGES ALLOWED WITHOUT WRITTEN APPROVAL FROM PIPE STRESS ENGINEER.

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--- EXISTING PIPE
--- NEW OR MODIFICATION OF EXISTING PIPE.

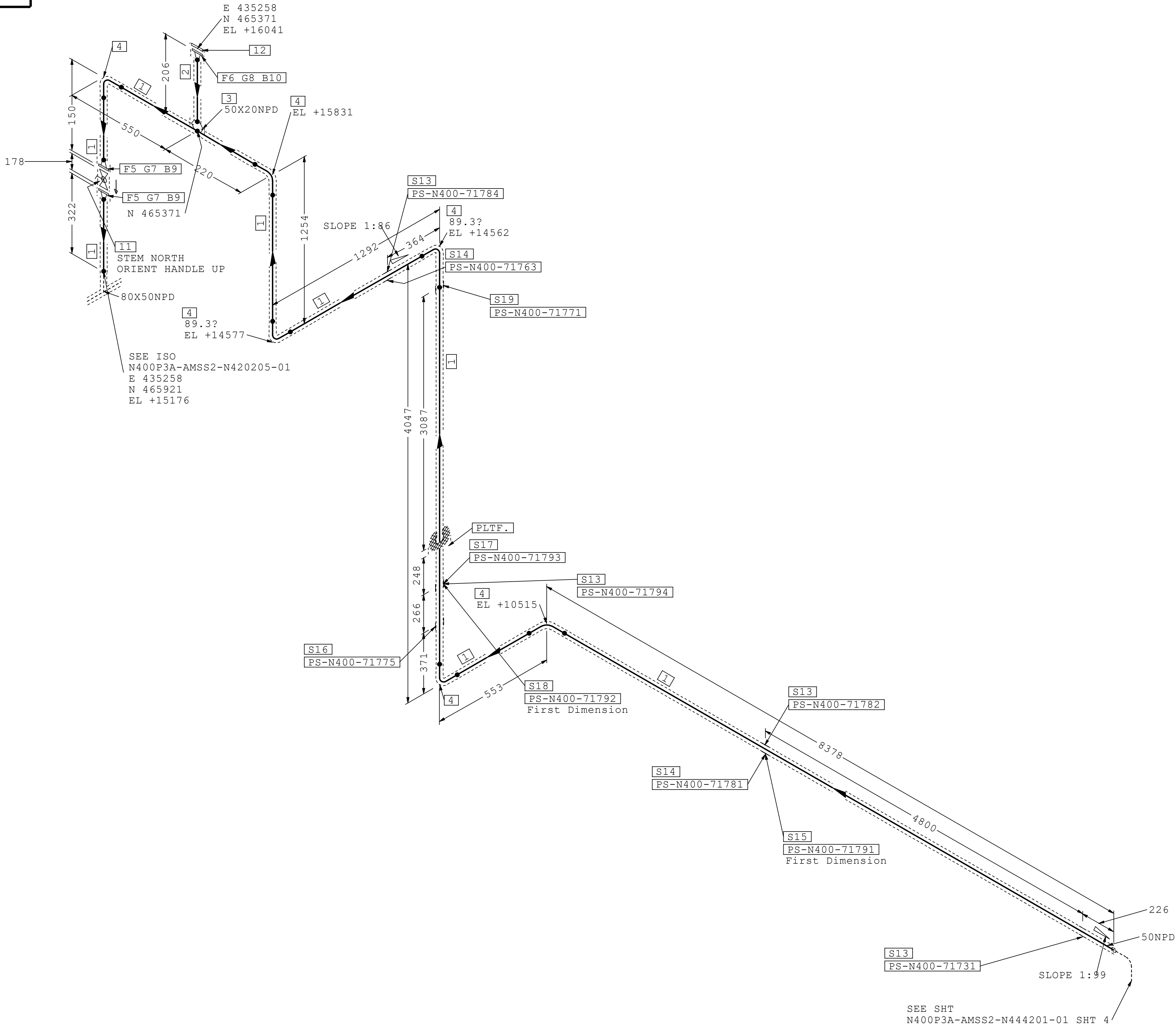
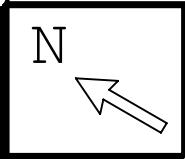
PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
1	Pipes (Length),ASME B36.19/B36.10,ASTM A312 Gr.TP316L,BE,Seamless,, S-40S, Undefined	50	CPPGKK29	PPPP21S40D30000000	8. 6 M
2	Ecc Reducer, ASME B16.9,ASTM A403 Gr.WP316L,BW Ends,Seamless,, S-40S, S-80S	50X40	CFHR0TC	DBRE20S5DP30010000	2
3	90Deg Elb LR, ASME B16.9,ASTM A403 Gr.WP316L,BW Ends,Seamless,, S-40S, S-40S	50	CDBGKLB	DBE920S5DP30010000	4
4	WN Flg,ASME B16.5,ASTM A182 Gr.F316L,RF/BW End,150 Lbs., Undefined, S-40S	50	CFGGTK6G	FGWN25S5SCH06L10000	3
5	WN Flg,ASME B16.5,ASTM A182 Gr.F316L,RF/BW End,300 Lbs., Undefined, S-80S	40	C26WFFJV	FGWN25S5SCH06L50000	2
INSTRUMENTS					
6	FVN444403	40			1
PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
7	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	50	C28Z7LJC	GKSW2100G00L100085400	4
8	SW Gk,ASME B16.20,RF as per ASME B16.5,-,300 Lbs,	40	C28Z7USN	GKSW2100G00L500085400	2
9	SBlt 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 100 mm Length	3/4	C10SN9ET	BTS222S00B010000	8
10	SBlt 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 90 mm Length	5/8	C10SN9ES	BTS222S00B010000	16
VALVES / IN-LINE ITEMS					
11	Bal Flg,FB,LP,ISO 17292,ASTM A351 Gr.CF3M,RF,150 Lbs., Undefined, Undefined	50	C26WDTVJ	VBAH80S4030AL1UY077MLE	1
INSTRUMENTS					
12	Flowmeter FTN444403	50			1
PIPE SUPPORTS					
13	SB08 A U10 441 0	50	..		1
14	CR01 2" 100 300 SS	50	..		3
15	WG02 2" C	50	..		2

0	2026.06.22	ISSUE FOR CONSTRUCTION		MLZ	ZLF	
REV	DATE	DESCRIPTION		DRAWN	CHECKED	APPROVED
BANC PANDA3 PROJECT						
蓝星安迪苏南京液体蛋氨酸装置技术改造项目						
		T.EN Chemical Engineering (Tianjin) Co.,LTD 德希尼布化学工程（天津）有限公司		 Bluestar Adisseo Nanjing Co.,LTD 蓝星安迪苏南京有限公司		
	PIPING ISOMETRIC					
UNIT	AREA		INSU. THK	STRESS CAT	DOCUMENT CLASS	
N400	N400P3A		30 mm		B	
LINE NUMBER					MARIAN SHEET	
80 AMSS2 N444201 A2DA10 PP					01	
PID NO.				SHEET	REV	
2069N-N400-PID-0021-N444				4 OF 5	0	

TPC S3D BANC2\BANC2\N400\2069N\Piping\Pipelines\2069N-N400-PID-0021-N444

N400P3A-AMSS2-N444201-01

N400P3A-AMSS2-N444201-01



NOTES: 1. FOR HOLD ITEMS, TPT PIPING FIELD ENGINEER TO CONFIRM FINAL DIMENSION AT SITE.
2. PIPE ROUTING AND SUPPORTS OF LINES DN40 AND SMALLER (NO STRESS ANALYSED) ARE TO BE CONFIRMED AT SITE BY PIPING SUB-CONTRACTOR.
3. FOR STRESS ANALYSED LINES, NO CHANGES ALLOWED WITHOUT WRITTEN APPROVAL FROM PIPE STRESS ENGINEER.

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--- EXISTING PIPE
— NEW OR MODIFICATION OF EXISTING PIPE.

PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
1	Pipes (Length),ASME B36.19/B36.10,ASTM A312 Gr.TP316L,BE,Seamless,, S-40S, Undefined	50	CPPGKK29	PPPP21S40D30000000	15. 8 M
2	Pipes (Length),ASME B36.19/B36.10,ASTM A312 Gr.TP316L,BE,Seamless,, S-80S, Undefined	20	C1GA7AUX	PPPP21S40D30000000	0. 1 M
3	Weldolet,MSS SP-97,ASTM A182 Gr.F316L,BW Ends,, S-40S, S-80S	50X20	C1GAAC33	LTLW0S5CH300000	1
4	90Deg Elb LR,ASME B16.9,ASTM A403 Gr.WP316L,BW Ends,Seamless,, S-40S, S-40S	50	CDBGKLJB	DBE920S5DP30010000	6
5	WN Flg,ASME B16.5,ASTM A182 Gr.F316L,RF/BW End,150 Lbs., Undefined, S-40S	50	CFGGTK6G	FGWN25S5CH06L10000	2
6	WN Flg,ASME B16.5,ASTM A182 Gr.F316L,RF/BW End,150 Lbs., Undefined, S-80S	20	CXSMBFK	FGWN25S5CH06L10000	1

PT NO	MESC NO & COMPONENT DESCRIPTION	DIAM. (NS)	IDENT CODE	COMMODITY CODE	QTY
7	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	50	C28Z7LJC	GKSW2100G00L100085400	2
8	SW Gk,ASME B16.20,RF as per ASME B16.5,-,150 Lbs,	20	C28Z7LJ3	GKSW2100G00L100085400	1
9	SBit 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 90 mm Length	5/8	C10SN9ES	BTS222S00B010000	8
10	SBit 2 HHx N,ASME B18.2.1/ASME B18.2.2,Full Length Threaded, - 70 mm Length	1/2	C10SN9DE	BTS222S00B010000	4

VALVES / IN-LINE ITEMS					
11	Bal Flg,FB,LP,ISO 17292,ASTM A351 Gr.CF3M,RF,150 Lbs., Undefined, Undefined	50	C26WDTVJ	VBAH80S4030AL1UY077MLE	1
12	Bld Flg,ASME B16.5,ASTM A182 Gr.F316L,RF,150 Lbs., Undefined, Undefined	20	CFG66DP7	FGBL25S5CH02L10000	1

PIPE SUPPORTS					
13	CR01 2" 100 300 SS	50	..		4
14	WG02 2" C	50	..		2
15	SB08 A U10 600 0	50	..		1
16	WE03 2" A 40S ASTM A312 Grade TP316L	50	..		1
17	WS02 2" B	50	..		1
18	SB08 A U10 370 0	50	..		1
19	WG04 2" A	50	..		1

0	2026.06.22	ISSUE FOR CONSTRUCTION	MLZ	ZLF	
REV	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED

BANC PANDA3 PROJECT
蓝星安迪苏南京液体蛋氨酸装置技术改造项目

	T.EN Chemical Engineering (Tianjin) Co.,LTD 德希尼布化学工程（天津）有限公司		Bluestar Adisseo Nanjing Co.,LTD 蓝星安迪苏南京有限公司
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PIPING ISOMETRIC					
UNIT	AREA		INSU. THK	STRESS CAT	DOCUMENT CLASS
N400	N400P3A		30 mm		B
LINE NUMBER					MARIAN SHEET
80 AMSS2 N444201 A2DA10 PP					01
PID NO.				SHEET	REV
2069N-N400-PID-0021-N444				5 OF 5	0

TPC S3D BANC2\BANC2\N400\2069N\Piping\Pipelines\2069N-N400-PID-0021-N444

N400P3A-AMSS2-N444201-01

N400P3A-AMSS2-N444201-01